COMPLETE ALGEBRA FORMULA SHEET

Essential Algebraic Identities, Rules, and Properties

1. Laws of Exponents & Logarithms

Category / Rule	Formula / Equation	Description / Notes
Product Rule	$a^m \times a^n = a^{m+n}$	Add exponents when multiplying same base
Quotient Rule	$a^m / a^n = a^{m-n}$	Subtract exponents when dividing
Power of a Power	$(a^m)^n = a^{m\times n}$	Multiply exponents
Negative Exponent	$a^{-n} = 1 / a^n$	Reciprocal with positive exponent
Log Product Rule	$\log_b(xy) = \log_b(x) + \log_b(y)$	Log of a product is sum of logs
Log Quotient Rule	$\log_b(x / y) = \log_b(x) - \log_b(y)$	Log of quotient is diff of logs
Log Power Rule	$\log_b(x^k) = k \cdot \log_b(x)$	Exponent becomes multiplier
Change of Base	$\log_b(x) = \log_c(x) / \log_c(b)$	Convert log base

2. Algebraic Identities & Factoring

Category / Rule	Formula / Equation	Description / Notes
Square of a Binomial	$(a + b)^2 = a^2 + 2ab + b^2$ $(a - b)^2 = a^2 - 2ab + b^2$	Expansion of binomial squares
Difference of Squares	$a^2 - b^2 = (a - b)(a + b)$	Product of sum and difference
Cube of a Binomial	$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$	Binomial expansions to power 3
Sum & Diff of Cubes	$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$	Factoring cubic polynomials
Trinomial Square	$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$	Square of three terms

3. Quadratic Equations & Progressions

Category / Rule	Formula / Equation	Description / Notes
Quadratic Formula	$x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$	Roots of $ax^2 + bx + c = 0$
Discriminant (Δ)	$\Delta = b^2 - 4ac$	>0: 2 real, =0: 1 real, <0: complex
Vertex Form	$f(x) = a(x - h)^2 + k$	Vertex is at point (h, k)
Arithmetic Seq (AP)	$a_n = a_1 + (n - 1)d$ $S_n = (n/2)[2a_1 + (n - 1)d]$	n-th term & sum of n terms
Geometric Seq (GP)	$a_n = a_1 \cdot r^{n-1}$ $S_n = a_1(1 - r^n) / (1 - r)$	n-th term & sum of n terms ($r \neq 1$)